

INDUSTRIAL DECARBONIZATION & CIRCULARITY

PRANGARA HANDBOOK

The Visual Guide to Tracking Carbon, Stopping Energy Leaks, Saving Crores, and Acing Global Compliance

⚡ Zero-LLM
Deterministic Math

💰 Cash-Positive
Decarbonization

IN Indian BEE MSME
Benchmarks

EU EU CBAM &
SEBI BRSR

Designed for factory owners, plant managers, engineers, auditors, and sustainability leaders.

PRANGARA OPERATING SYSTEM · VERSION 2.0 · AUTONOMOUS AGENTIC EDITION

1. What is Prangara? (Explain Like I'm 5)

🦊 Think of Prangara as a "Smart Health Tracker & Doctor" for a Giant Factory:

Just like your smartwatch counts your steps, detects when you didn't sleep well, and tells you what healthy foods will give you energy, **Prangara** tracks a factory's electricity, coal, diesel, raw materials, and truck deliveries. It spots where money is secretly leaking out of hot steam pipes and old motors, and shows the owner how to save millions of rupees while keeping the air clean!

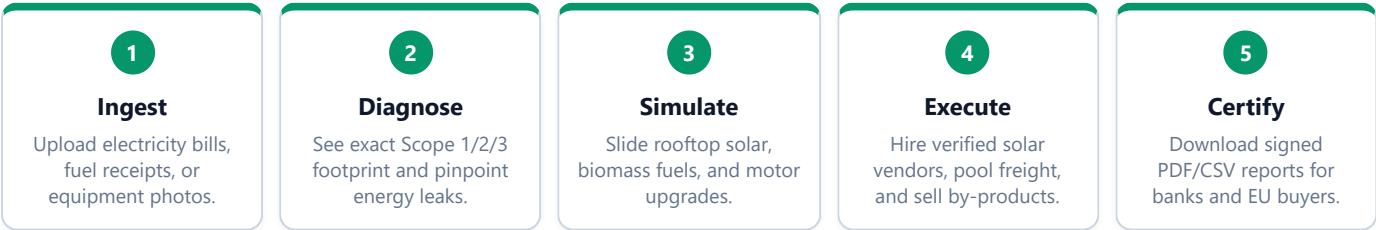
❌ Without Prangara

- **Costly Auditors:** Paying ₹5L–₹25L to consulting firms for 4-month audits.
- **Hidden Leaks:** Old IE1 motors and uninsulated steam pipes wasting ₹20L/yr.
- **CBAM Penalty Fear:** High carbon export taxes deducted at European ports.
- **Empty Trucks:** Paying full price for half-empty transport trucks.
- **Paying to Dump Waste:** Dumping ash and cotton scraps into landfills.

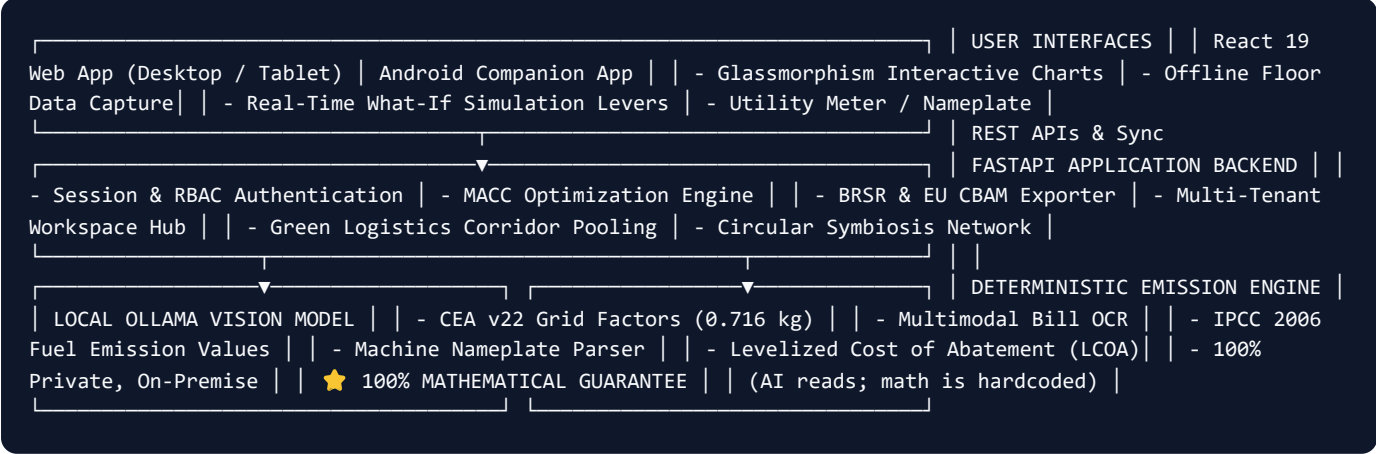
✅ With Prangara

- **Instant Baseline:** Snap pictures of bills/nameplates for instant audit grade.
- **Automated Leak Detection:** Spots efficiency gaps compared to peer mills.
- **Export Ready:** 1-click verified reports for EU CBAM customs and SEBI BRSR.
- **Smart Cargo Pooling:** Shares container trucks with neighbors to cut bills by 30%.
- **Circular Exchange:** Sells waste ash/scraps to other factories as raw material.

2. The 5-Step Factory Journey



3. System Architecture & Zero-LLM Math



4. Complete Page-by-Page Walkthrough

1. Workspace Hub

/WORKSPACE

What it is: Control center for managing one or multiple factories under your organization.

Features: Multi-plant comparison cards, machinery registries (boilers, motors, chillers), intake history, and auditable data quality scores.

How to use: Register new plant locations, switch your active factory, or drill down into specific equipment logs.

2. Account & Roles

/ACCOUNT

What it is: User identity, security settings, and organization permissions.

Features: Role-Based Access (Factory Owner, Plant Engineer, Auditor, Vendor), company switcher, and API tokens for ERP integration.

How to use: Invite team members with role-specific access, manage personal profile, and configure webhooks.

3. Notifications

/NOTIFICATIONS

What it is: Factory early-warning and operational inbox.

Features: Sudden energy spike warnings (e.g. boiler fuel up 14%), incoming quotes from solar contractors, and CBAM deadline alerts.

How to use: Check pending alerts daily and click on an item to jump straight to the offending machine or bid.

4. Overview Dashboard

/OVERVIEW

What it is: Single-page executive summary for the CEO and board.

Features: "Cash-Positive Opportunity" hero banner (₹ savings/yr), Scope 1/2/3 summary cards, and regional peer benchmark standing.

How to use: Review your headline emissions, potential rupee profit from greening, and top 3 priority actions.

5. Plant Data Intake

/ASSESSMENT

What it is: Guided portal to input your factory's numbers with zero headache.

Features: 8-step wizard, local AI bill/nameplate camera reader, spreadsheet bulk uploader, and evidence attachment.

How to use: Upload monthly electricity bills or fuel sheets; review the auto-extracted values and click "Run Baseline".

6. Emissions Footprint

/FOOTPRINT

What it is: Detailed forensic inventory of your total greenhouse gases.

Features: Scope 1 (boilers/diesel), Scope 2 (grid electricity), Scope 3 (freight/waste), and interactive Sankey flow chart.

How to use: Inspect emissions by stream. Click any row to view the official CEA/IPCC emission factor and audit trail.

7. Leak Points Hotspots

/LEAKS

What it is: Automated energy detective spotting wasteful machines.

Features: Anomaly detection (e.g. old IE1 motors, uninsulated steam pipes, air leaks) ranked by severity (Critical / High).

How to use: Plant engineers review this page before shutdowns to prioritize maintenance and equipment upgrades.

8. What-If Simulator

/SCENARIOS

What it is: Interactive simulation sandbox to test green upgrades.

Features: Sliders for Rooftop Solar (0-100%), Boiler Fuel Switch (Coal → Biomass), Recycled Material %, and IE4 Motors + VFDs.

How to use: Drag sliders to see real-time recalculations of carbon reduction, annual ₹ savings, and payback period.

9. Circular Actions

/ACTIONS

What it is: Curated catalog of pre-costed decarbonization projects.

Features: Categorized into Quick Wins (payback < 12 mos), Cash-Positive, and Net-Cost. Shows Capex, Opex delta, and carbon eliminated.

How to use: Filter by Quick Wins and push projects directly to the Marketplace to request contractor quotes.

10. Abatement Portfolio

/PORTFOLIO

What it is: The Marginal Abatement Cost Curve (MACC) prioritization engine.

Features: Interactive MACC chart showing projects that make money (below zero line) vs projects requiring net capital.

How to use: Present this curve to CFOs and banks to prove that early quick-win profits fund larger future green upgrades.

11. Marketplace & RFQs

/MARKETPLACE

What it is: B2B clean tech directory and Request-For-Quote procurement hub.

Features: Verified solar installers, ESCO auditors, low-carbon materials catalog, and side-by-side bid comparison matrix.

How to use: Send project specs to 3 regional contractors in 1 click, compare prices and warranties, and accept the winning quote.

12. Green Logistics

/LOGISTICS

What it is: Industrial corridor freight optimization and cargo pooling engine.

Features: Multi-factory truck pooling (milk runs), modal comparison (Diesel Truck vs EV vs Train), and major export corridor routes.

How to use: Enter shipment weight and click "Join Cargo Pool" to share empty truck space, slashing freight costs and emissions.

13. Circular Network

/CIRCULAR-NETWORK

What it is: Industrial symbiosis exchange: one plant's waste is another's input.

Features: Marketplace for by-products: Fly Ash, Cotton Comber Noil, Spent Solvents, Slag, and Bagasse with avoided carbon metrics.

How to use: Monetize production scraps by selling to nearby buyers or purchase secondary materials at 30% discount.

14. Compliance & CBAM

/COMPLIANCE

What it is: Regulatory compliance evaluator and automated reporting tool.

Features: EU CBAM indicative tax exposure calculator, SEBI BRSR Core readiness checks, and 1-click PDF audit sheet export.

How to use: Calculate export tax liabilities before shipment and export signed audit packets for customs and banks.

15. Methodology & Emission Factors

/METHODOLOGY

What it is: The transparent, open mathematical ledger and factor library powering the whole platform.

Features: Official Central Electricity Authority (CEA) grid baseline v22 factors, IPCC 2006/2019 fuel tables, and mathematical formulas for LCOA.

How to use: Share this page with certified auditors and lenders to verify calculations with zero black-box obscurity.

5. Official Indian Industrial Benchmarks (BEE MSME Study)

Prangara automatically grades your factory against real cluster medians from the Bureau of Energy Efficiency:

Sector	Key Industrial Clusters	Thermal (GJ/t)	Elec (kWh/t)	Baseline (\$tCO ₂ e/t\$)
Secondary Steel Re-Rolling	Mandi Gobindgarh (PB), Jalna (MH), Bhavnagar (GJ)	1.85 (1.4–2.6)	110 (80–160)	0.22
Aluminium Die-Casting	Pune (MH), Coimbatore (TN), Ahmedabad (GJ)	5.80 (4.2–8.5)	620 (450–950)	0.94
Textile Processing & Dyeing	Surat (GJ), Tirupur (TN), Panipat (HR)	18.50 (12–28)	520 (350–800)	2.15
Specialty Chemicals & Dyes	Ankleshwar (GJ), Vapi (GJ), Tarapur (MH)	14.20 (8–22)	880 (500–1400)	1.85
Ceramics & Vitrified Tiles	Morbi (GJ), Thangadh (GJ), Khurja (UP)	6.40 (4.8–9.2)	180 (120–260)	0.51
Foundry & Castings	Rajkot (GJ), Belgaum (KA), Coimbatore (TN)	4.90 (3.5–7.2)	680 (520–980)	0.88
Paper & Kraft Packaging	Vapi (GJ), Muzaffarnagar (UP), Morbi (GJ)	10.20 (7.5–15)	360 (280–520)	1.12
Plastics Injection Molding	Daman (UT), Ahmedabad (GJ), Peenya (KA)	1.10 (0.5–2.2)	950 (600–1600)	0.74

6. A Real-Life 30-Day Story: Rajesh Garments (Tirupur)

- Day 1: The Urgent Crisis**
Rajesh runs a knitwear dyeing plant in Tirupur. A major European apparel client warns him that without a certified carbon disclosure, his export goods will face severe CBAM import tax deductions at Hamburg port.
- Day 3: Ingestion in 15 Minutes**
Rajesh uploads his electricity bills and diesel generator logbooks using the Plant Data Intake. Local Ollama AI extracts all numbers without sharing sensitive company files. Total baseline: **2,150 \$tCO₂e\$/year**.
- Day 7: Identifying the Waste**
The Leak Points page reveals his wood-fired boiler operates at only 61% thermal efficiency, and outdated IE1 motors are burning ₹11 Lakhs in excess electricity annually.
- Day 12: Simulating the Fix**
In the What-If Simulator, Rajesh tests **40% Rooftop Solar**, **Biomass Briquettes**, and **IE4 Motors**. The model reveals: **36% carbon drop, ₹38.5 Lakhs saved annually, and full payback in 2.4 years!**
- Day 20: Sourcing & Logistics**
He sends an RFQ to 3 verified solar installers on the Marketplace and joins a shared container truck pool on Green Logistics, saving 28% on export freight to Chennai Port.

Day 30: Certified & Compliant

Rajesh exports the official signed PDF report from the Compliance module. His European client approves the shipment with zero tax penalties and renews his supply contract for 3 years.

7. Quick Reference & Shortcuts

Action	Location / Shortcut	Notes
Start Both Servers	Double-click start.bat in root	Launches FastAPI backend (8000) & Vite frontend (5173)
Stop Both Servers	Double-click stop.bat in root	Cleanly terminates both background tasks
Global Command Palette	Press Ctrl + K anywhere	Jump instantly to any page, machine, or report
Ask Prangara AI	Click ✨ Sparkles icon	Answers questions with citations from your plant data